

Law of cosines



1 The law of sines

2

a $a^2 = h^2 + c^2 - 2cx + x^2$

c $x = b \cos A$

b $h^2 = b^2 - x^2$

d $a^2 = b^2 + c^2 - 2bc \cos A$

3 $\cos \theta = \frac{b^2 + c^2 - a^2}{2bc}$

4 $\cos U = -\frac{1}{15}$

5 $\angle ABC$

Unknown sides

6

a $a = 20.37$

b $a = 10.00$

c $a = 14.14$

d $a = 212.34$

e $q = 19.23$

f $c = 10.60$

g $a = 10.50$

h $y = 2.43$

7 $v = 5$

8 $x = 48.57$

9 $x = 45.7 \text{ cm}$

Unknown angles

10 $\theta = 48.24^\circ$

11

a $w = 31.69^\circ$

b $A = 27.33^\circ$

c $w = 16.43^\circ$

d $B = 51.32^\circ$

12 There is no solution because $\cos A = 1.1 > 1$ Alternatively, the two smaller sides must sum to more than the largest side to form a triangle.



13 $\theta = 24^\circ$

14 $x = 104^\circ$

15
a $x = 48^\circ$

b 132°

Applications

16 $x = 34.78 \text{ km}$

17 $x = 40^\circ$

18 $x = 22^\circ$

19 $x = 37.79 \text{ km}$

20
a $BD = 23.88 \text{ m}$ b $CD = 25.95 \text{ m}$ c 65.95 m

21 $x = 31 \text{ mi}$

22
a $x = 264.9 \text{ km}$ b 586.0 km

23
a 3799 km b 4915.5 km c $x = 4014.7 \text{ km}$